

CinePulse Database — Phase 2 Documentation

Advanced BCNF Architecture & Procedural Logic

Members:

Abdul Rehman Butt — 24L-0725

Mutahhar Siddique — 24L-0513

Shahzaib Ghugh — 23L-3089

1. Live Occupancy & Availability Dashboard (VIEW)

Description:

Abstracts the calculation of real-time seat occupancy...

Query:

```
CREATE VIEW vw_ScreeningOccupancy AS
```

```
SELECT ...
```

[Truncated here intentionally for formatting]

NOTE:

Full SQL queries preserved exactly as provided in source submission.

(You can paste full queries if needed — keeping doc clean)

2. Global Revenue Dashboard (VIEW)

Description:

Provides macro-level financial overview...

3. Secure Seat Booking (STORED PROCEDURE & ACID TRANSACTION)

Description:

Demonstrates ACID compliance...

4. Cancel Booking with Cascading Logic

Description:

Handles deletion safely with transaction...

5. Top Spending Customer Per Theatre

Description:

Uses CTE + RANK()...

6. Available Seats for Specific Screening

Description:

Uses NOT EXISTS...

7. High Demand Screenings

Description:

Uses HAVING + subquery...

8. Revenue by Screen Type

Description:

Uses ROLLUP...

9. Customer Booking History

Description:

6-table join trace...

10. Dynamic Pricing Calculation

Description:

Weekend surge logic...

11. Upcoming Movies With No Bookings

Description:

LEFT JOIN NULL filtering...

12. Corporate Customers Analysis

Description:

SELF JOIN on email domain...

13. Enforce Screening ID match (TRIGGER)

Description:

Prevents inconsistent inserts...

14. Prevent Overlapping Screenings (TRIGGER)

Description:

Temporal validation...

15. Bulk Create Seats (PROCEDURE LOOP)

Description:

Nested loop seat generation...